

Interactive exploration of benchmarking results with *bettr*

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Summary

Method benchmarking is a core part of many research fields and aims to establish best practices in method selection and application as well as help identifying gaps and possibilities for improvement in existing methods. A typical benchmarking study scores a set of methods using a variety of different metrics intended to capture different aspects of performance and usability (e.g., area under the receiver operating characteristic (ROC) curve for a classification task, or execution time). For practical purposes, e.g., when a user needs to select a method for their own analysis, it is often the case that not all of the evaluated metrics are equally important. For example, the ability of a statistical method to handle arbitrarily complex experimental designs may not be relevant if the user only needs to perform a two-group comparison, while memory frugality may be of high importance if the analysis will be performed on a system with limited computational resources. Hence, having the ability to easily and adaptively create a weighted summary score by which the methods can be ranked, or alternatively to create a visual summary of only the desired metrics, would make it possible for users to tailor their method choice based on the aspects that are most relevant to them.

Inspired by the OECD 'Better Life Index', which allows users to score OECD countries by a customizable weighted average of various factors contributing to human well-being, we developed the *bettr* R/Bioconductor package to provide similar support for consumers of benchmarking studies. Given a table with values of evaluation metrics across the evaluated methods, as well as optional method annotations and metric groupings, *bettr* allows users to visualize performance summaries emphasizing the aspects and evaluation metrics that are most important to them (Figure 1). *bettr* can be used interactively as an R/Shiny application (Chang et al., 2025), or programmatically by calling the underlying functions directly for full reproducibility and integration into analytical pipelines (see the package vignettes for more details).

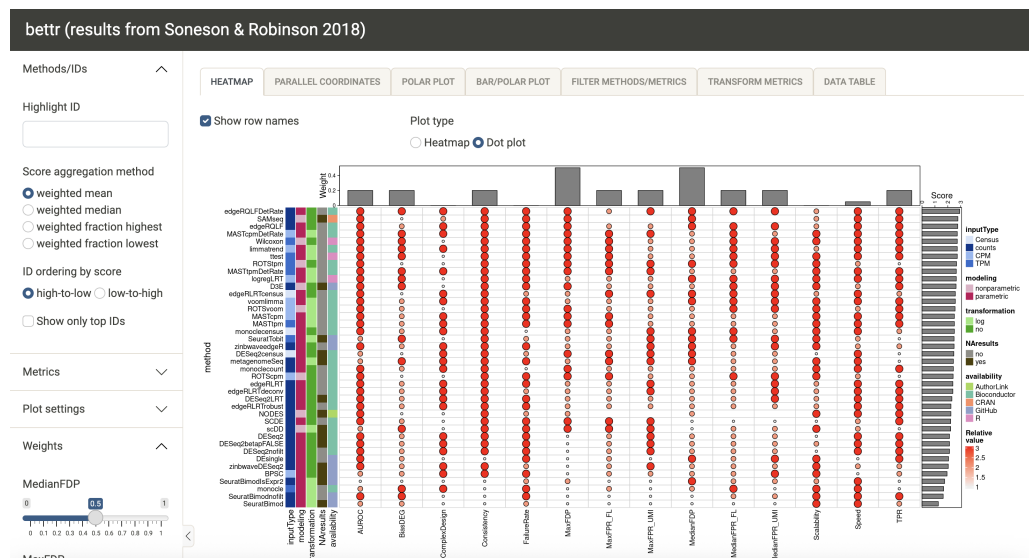


Figure 1: Screenshot of a *bettr* application illustrating the benchmarking results from Sonesson & Robinson (2018). The size and color of the circles indicate the evaluation score for each method (rows) and metric (columns), with higher values being more favorable. The bars above the plot indicate the current weights for the metrics (these are controlled by the sliders in the bottom of the left sidebar), and the bars to the right of the plot represent the aggregated scores for the methods (in this case, the weighted mean across metrics). The colored bars to the left of the plot represent categorical annotations and characterizations of the methods, which are not taken into account when determining the ranking. In addition to the displayed dot plot, the results can be visualized in several other ways, including a heatmap, polar plots, and using parallel coordinates (horizontal tabs). Moreover, the set of methods and metrics to include in the display can be controlled by the user via the 'Filter methods/metrics' tab, and the transformations applied to the different metrics can be controlled via the 'Transform metrics' tab. Finally, the data can be displayed in tabular form and exported via the 'Data table' tab. In addition to providing controls for the metric weights, the left-hand side bar allows the user to control how metric scores are aggregated to form the final ranking, and how metrics are grouped together.

Statement of Need

Method benchmarking is a prolific area of research across different fields of application. However, in many cases the conclusions presented in benchmarking papers are hard for a reader to explore further, since the results (e.g., performance metrics) are often not readily available or provided in an explorable format (Sonrel et al., 2023). As a consequence, much of the interpretation is left solely to the benchmarker and the view that they choose to highlight in the static paper figures. *bettr* aims to address this asymmetry and enable both authors and readers of benchmarking studies to explore the results more easily and flexibly. This has the potential to increase the usefulness of published benchmarks by democratizing the interpretation of the benchmark results and enabling a reader to select the most suitable method for their specific setup in a more nuanced manner.

State of the Field

Most published visualizations of benchmark results are created with custom scripts using generic R or python graphics libraries. A small number of dedicated viewers for benchmark results exist, and *bettr* was designed to complement their functionality, rather than to replace them. The funkyheatmap R package (Cannoodt et al., 2025) generates elaborate, publication-ready static heatmaps of diverse types of benchmark results. It is also wrapped into a standalone executable as well as a [nextflow module](#), and the funkyheatmap.js [extension](#) can be used

to create visualizations that can be interactively browsed and sorted by a selected metric. However, it was not designed to accommodate in-depth exploration via interactive aggregation or re-weighting of metrics by the user, highlighting a fundamental difference in design philosophy compared to *bettr*. Benchmarking platforms such as [OpenEBench](#) ([Capella-Gutierrez et al., 2017](#)) and [OpenProblems](#) ([Luecken et al., 2025](#)) also produce result visualizations for the included benchmarks, with OpenProblems using funkyheatmap for this purpose. These again serve a different use case than *bettr*: by being fully integrated into the respective platforms, they provide consistent visualizations for all included benchmarks, but are not as easily accessible to a user aiming to visualize the results of their own study. Similarly to funkyheatmap, they also do not support user-specified metric weighting or aggregation.

Thus, to the best of our knowledge, *bettr* is the first generic tool for benchmark visualization that combines interactivity and flexibility in metric weighting with ease of use and a transparent, programmatic interface.

Software Design

The design philosophy behind *bettr* focuses on flexibility, accessibility, and reproducibility. By supporting several input formats, from collections of data frames via a single SummarizedExperiment object ([Morgan et al., 2025](#)) to an all-encompassing JSON file, *bettr* can be used for downstream analysis and visualization of results generated using a wide range of benchmark setups and programming languages. For example, a user working locally in R may find it more convenient to generate a set of data frames, while an automated benchmarking workflow could export all necessary components in a single, platform-independent JSON file. *bettr* can also be deployed in server mode, allowing users to upload their own input data (in JSON format) to a running app. *bettr* is fully open source and easily installable as an R package, mainly distributed via Bioconductor with the most recent development version also available on GitHub. In addition, it is available via [r-universe](#), which among other things provides binaries for several platforms, including WebAssembly/WebR; (<https://github.com/r-wasm/webr/>). As a complement to the interactive interface, full reproducibility is enabled by the equivalent programmatic interface to the functionality, as well as the ability to export the processed data as either a shareable csv file containing the metric values as well as the final score, or an R list that can be used directly as the input for further analysis and visualization.

Research Impact Statement

In 2025, *bettr* was [downloaded](#) almost 3,000 times (by 1,593 unique IPs) from Bioconductor alone. Adoption of *bettr* is expected to increase further in the near future, as it has become the first supported integration for automated metrics reporting within the [Omnibenchmark](#) project. Omnibenchmark ([Mallona et al., 2026](#)) is a benchmarking system that automates and standardizes routine aspects of benchmarking through standardization and formalization of benchmarking plans. During execution, Omnibenchmark collects computational performance metrics (e.g., peak memory usage, CPU utilization, run time, etc), as well as, where applicable, algorithmic performance metrics (e.g., F1 scores, ARI, etc). These results are then exported via a command-line interface in a JSON format compatible with *bettr*, enabling automated reporting. This integration lowers the barrier for benchmark authors to produce rich, interactive summaries of complex benchmarking studies without requiring custom visualization pipelines. Hence, broader adoption of Omnibenchmark as a benchmarking framework is expected to further drive the use of *bettr* as an interactive platform for exploring and interpreting benchmark results.

Availability and Examples

bettr is available via [Bioconductor](#) and on [GitHub](#). A collection of example data sets and *bettr* configurations are available from <https://github.com/csoneson/bettr-examples>. In addition, an example instance, using data from Soneson & Robinson (2018), is deployed on <https://csoneson.shinyapps.io/soneson2018de/>.

AI Usage Disclosure

The majority of the *bettr* code base was developed over several years, in a public GitHub repository, without the use of generative AI. For one pull request (#25), adding the capabilities to support JSON files as input and to cache the state of the app, Claude Sonnet 4.5 (Anthropic) was used to assist with drafting implementation of new functions and UI components based on predefined design documents, while maintaining existing functionality as is. At least two of the package developers reviewed the code carefully, edited it where necessary, and verified that the new code performed as intended and did not introduce regressions in existing functionality (which was further verified using the comprehensive set of existing unit tests).

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